HANDS-ON · LENSES & TELESCOPES

Light Buckets: Build a Telescope

Name _____

Work through the **Light Buckets** page one step at a time. Each step on the page has a green **"With your lenses"** box. Do what it says with your real lenses, then fill in the same step here.

Before you start

- Put tape on the edge of the **wider** lens and write **A**. The other is **B**.
- Hold lenses by their **edges**; fingerprints blur everything.
- Kit: 2 lenses, meter stick, white index card, pencil.

Safety: never look at the Sun through a lens or a telescope, and never use a lens to focus sunlight on anything.

1 Light travels in straight lines

Light from something very far away, like a star, reaches you as rays that are _____ (side by side).
On the page, the top and bottom rays differed by less than 1° once the bulb was about _____ away.

2 Light bends TRY IT WITH A LENS

Hold Lens A about **2 cm** above this sentence, then slowly lift it toward your eye: **Galileo saw four moons of Jupiter in 1610.**

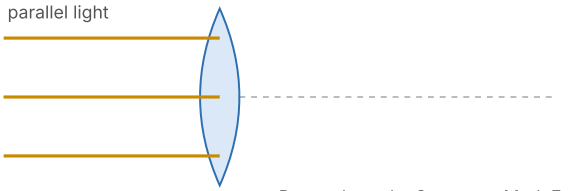
At 2 cm the words look: ☐ bigger ☐ smaller ☐ the same. Lifted higher they become: ☐ blurry ☐ upside down ☐ both

On the page, which material bent light the most? _____ Light coming **out** of glass gets trapped past _____ °.

3 Shapes LOOK, FEEL, MEASURE

LENS	WIDTH (MM)	MIDDLE FATTER OR THINNER?	CONVERGING OR DIVERGING?
A			
B			

Which lens bulges more? _____



Draw where the 3 rays go. Mark F.

4 The focal point PREDICT

I predict Lens _____ has the **shorter** focal length, because _____

On the page: a glass lens with f = 20 cm, switched to water, has f = _____ cm. To get back to 20 cm, the water lens had to be ☐ more curved ☐ less curved.

5 Lenses make pictures TRY IT WITH A LENS

Hold Lens A toward a bright window, 3 m or more away. Start with the card touching the lens and slowly pull it back until the picture is sharp.
The picture is ☐ right-side up ☐ upside down.
Bonus: with a lamp only 1 m away, the sharp spot is ☐ closer to ☐ farther from the lens.

sketch the picture on your card

On the page (f = 10 cm): same-size picture with the candle at _____ cm and the picture at _____ cm. To make the picture **bigger**, move the candle ☐ closer ☐ farther. Candle closer than F: _____

6 Measure the focal lengths DO IT

Practice: the page's mystery lens had f = _____ cm. With the window only 1 m away, its sharp spot was at _____ cm, ☐ longer ☐ shorter than f.

Your lenses: stand 3 m or more from a bright window. Focus the window on the card. Measure from the **middle of the lens** to the **card**. Do it twice.

LENS	TRY 1 (CM)	TRY 2 (CM)	AVERAGE F (CM)
A			
B			

Was your Step 4 prediction right? ☐ yes ☐ no Type your numbers into the page (Step 6) so it can use your lenses.

7 A magnifying glass TRY IT WITH BOTH LENSES

Galileo saw four moons of Jupiter in January 1610.
The Moon is about 384,000 km away from Earth.
Light from the Sun takes about 8 minutes to reach us.
The Andromeda galaxy is 2.5 million light-years away.

Hold the lens about 2 cm from your eye. Bring this text up until it's sharp.

	SMALLEST LINE (1–4)	LENS TO PAPER (CM)	POWER = 25 ÷ F
A			×
B			×

The better magnifier is Lens _____, the one with the ☐ shorter ☐ longer f. The lens-to-paper distance was ☐ less ☐ more than f.

8 Build a telescope

8a · Plan

Objective (front) = the **longer** focal length	Lens _____ f_o = _____ cm
Eyepiece (at your eye) = the **shorter** focal length	Lens _____ f_e = _____ cm
Predicted spacing: $f_o + f_e$	_____ + _____ = _____ cm
Predicted magnification: $f_o \div f_e$	_____ ÷ _____ = _____ ×

8b · Build. Objective at 0 cm on the meter stick, pointed at the poster. Eyepiece at your predicted spacing, eye 1–2 cm behind it. Line up eye, both lenses and poster. Slide the eyepiece slowly until the poster is sharp.

8c · Spacing when it's sharp	_____ cm (difference from prediction: _____ cm)
8d · The view is…	☐ upside down ☐ right-side up · ☐ bigger ☐ smaller
8e · Swap the lenses (short f in front). Things look…	☐ bigger ☐ smaller · $f_e \div f_o$ = _____ ×
On the page · Jupiter through my telescope	I could see _____ moons at _____ ×

your view

Sketch what you see through your telescope. Which line of the poster could you read **with** the telescope but not **without**?

_____

9 Light buckets

Your pupil is about 7 mm wide in the dark. My objective is _____ mm wide (from Step 3).
Light collected vs. my eye: `(width ÷ 7)²` = (_____ ÷ 7)² = (_____)² = _____ times as much light.
On the page, the smallest opening that showed more than 1,000 stars was about _____ mm.
A telescope twice as wide collects _____ times as much light, because

_____

10 Lenses vs. mirrors LOOK CLOSELY

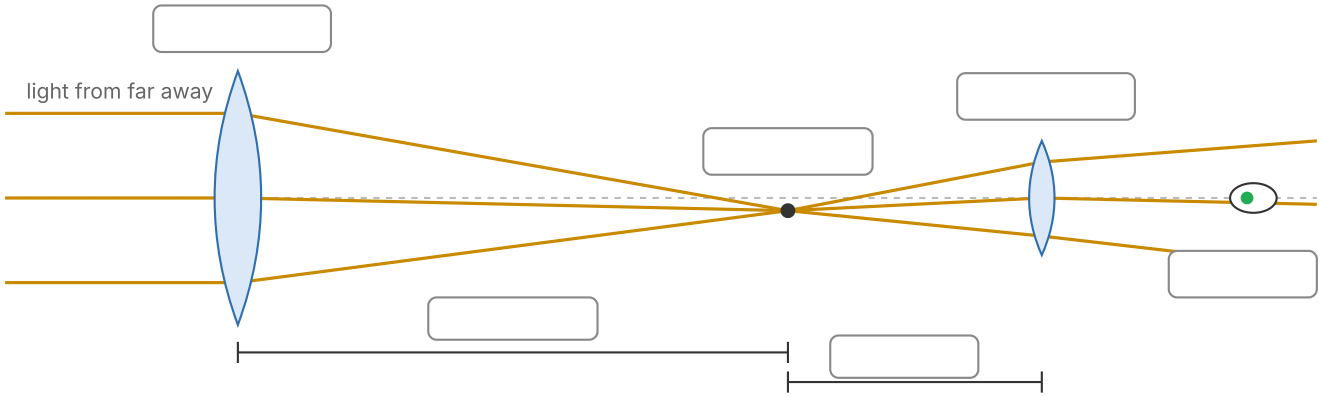
Through my telescope, a bright edge has colored fringes: ☐ yes ☐ a little ☐ no
Two reasons the biggest telescopes use mirrors instead of lenses:

_____

_____

★ Put it together: label the telescope

Write each word in the right box: **objective · eyepiece · image · f_o · f_e · eye**



? Think about it

Galileo's best telescope magnified about 20×. Using what you learned, what would you change about **your** telescope to make it 20×? Which lens would you replace, and with what focal length?

_____

_____

Exit ticket. In your own words: what does the **objective** do, and what does the **eyepiece** do? Why do astronomers want **wide** telescopes?

_____

_____

_____

INSTRUCTOR · ANSWER KEY

Light Buckets: answer key

Student focal lengths depend on the lenses handed out; check that their numbers are consistent rather than "right". Careful by-eye measurements land within about 10% of the true focal length. Fill in your kit's values in the last column before the event.

STEP	EXPECTED ANSWERS	YOUR KIT
1	**Parallel.** Page: under 1° once the bulb is about 3 m away (from a 5 cm lens).	
2	At 2 cm: **bigger** (magnifier). Lifted: **blurry**, then **upside down** (past the focal length). Diamond bends most (light slows most, n = 2.42). Trapped in glass past **42°** (41.8°).	
3	Both lenses **fatter in the middle = converging**. Rays drawing: all three meet at one point on the axis (F), then spread out again.	Width A: ____ mm Width B: ____ mm
4	The lens that **bulges more** has the shorter f (bends light more). Page: water f ≈ **30 cm**; the water lens must be **more curved**.	Shorter f: Lens ___
5	Picture **upside down**. Bonus: a nearer lamp focuses **farther** from the lens. Page: same size at **20 cm / 20 cm** (twice f); bigger picture: candle **closer** (between 10 and 20 cm); closer than F: **no picture** on the paper (magnifier instead).	
6	Mystery f varies. With the window 1 m away, the sharp spot is **longer** than f. Two tries should agree within ~1–2 cm.	f_A: ____ cm f_B: ____ cm
7	The **shorter**-f lens magnifies more (≈ 25 ÷ f) and reads a smaller line. Lens-to-paper distance is **less** than f. That lens becomes the eyepiece.	Power A: ____× Power B: ____×
8	Objective = longer f; eyepiece = shorter f. Spacing ≈ $f_o + f_e$ (the sharp spot is often a little shorter: the eye can focus on a nearby image). View is **upside down** and **bigger**. Swapped: $f_e \div f_o < 1$, things look **smaller**. Jupiter on the page: moons separate from about 3×; 2–4 moons visible depending on power. Example: f_o = 30 cm, f_e = 6 cm → 36 cm apart, 5×; swapped 0.2×.	Spacing: ____ cm Power: ____×
9	Example: 40 mm → $(40 \div 7)^2 = 5.7^2 \approx 33\times$. Page: just over 1,000 stars at about 0.8 m. Twice as wide → **4×** the light, because light is collected over the **area**, which grows with width squared.	Light: ____×
10	Simple lenses often show red/blue fringes. Mirrors: no color fringes; supported from behind so they don't sag; light only touches the front surface; can be built from segments (the 39 m ELT has 798).	
★	Left to right: **objective** (front lens), **image** (where the rays cross), **eyepiece**, **eye**; bottom brackets: $\mathbf{f_o}$ (objective to image) and $\mathbf{f_e}$ (image to eyepiece).	
?	Magnification = $f_o \div f_e$, so use a **longer-f objective** or a **shorter-f eyepiece**. E.g. keep f_o = 30 cm and use f_e = 1.5 cm, or keep f_e = 6 cm and use f_o = 120 cm (a 126 cm long telescope!).	
Exit	The objective collects light and makes a small image of a faraway object; the eyepiece magnifies that image like a magnifying glass. Wider telescopes collect more light, so they show fainter things.	

Facilitation tips. Step 6 works best with a bright window across the room; a ceiling light overhead is hard to use. Partners help: one holds the lens and card, one measures. In Step 8 the common problems are lenses swapped, lenses not lined up with the eye, the eye too far from the eyepiece, or sliding past the narrow sharp spot too fast. Tape lens labels on before handing out. If a lens has f longer than ~60 cm, have students focus onto a wall instead of a hand-held card.